Execution and Scoring Semantics in the Public AutumnBench Release: A Bounded Audit

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Scope. This independent note combines saved CPU-only pilot results with a review of the published task definition and pinned public code. It documents a protocol/scoring discrepancy in the inspected release; it does not assess the validity of the paper's reported experiments.

A. Context and research question

WorldTest evaluates world-model learning through behavior in derived challenge environments, rather than only prediction on observed trajectories [1, §4.1]. Such evaluation requires the task definition, interaction protocol, simulator transitions and score to refer to the same query. A historical observation index, an interface action count and a latent guard event need not identify the same answer.

This audit asks whether the public AutumnBench Change Detection (CD) interface preserves the meaning of the reported change time through frame selection and submission. Paper statements, saved runtime observations, static code findings and source-derived reconstructions are distinguished below.

B. Verified executable baseline and its boundary

The unchanged native interpreter compiled and imported with Python 3.11/GCC12 in an isolated CPU environment. The frozen public dataset contains **20 environments and 60 tasks** [4], compared with **43 environments and 129 tasks** in the paper [1]. Scientific execution was limited to `27VWC`, seed 0.

Masked frame prediction (MFP). Native replay matched **117/117** released frames at all visible cells and selected the released correct option. Official positive and negative scoring checks passed, with raw rewards +1 and −1, respectively. These are interpreter/scorer checks, not learned-model predictions.

Planning. A supplied MFP action trajectory served as a released planning witness. The official planning environment reached its target after **111 executed steps**, returning reward +1. The input witness contained 116 actions; the plan was not independently discovered and no optimality claim is made.

Change detection. The saved guard first triggered at time 6 after six refill clicks; a subsequent no-op ended at time 7. The harness declared a change, chose frame 5, then submitted. Its `reward == 1 and terminal` assertion failed. **The returned numerical CD reward was not retained**; it cannot be reported as an observed value.

Limits. The original CD harness omitted the official initial observation and did not persist its final return before asserting. An earlier shallow-planning gate was corrected by a preserved protocol amendment; both failed attempts remain retained. The second pilot environment, `7WWW9`, and seed 1729 were not completed. **Zero controlled program mutations and zero learned-world-model evaluations** were performed. The complete multi-family baseline was not established.

Frozen source references.

Autumn.cpp [2]: `0929d175c3d296de091ac4b998d09b9b55a38f67`

MARAProtocol [3]: `d0d4e9251151778415cc5195c5dfb8296a5ded24`

Public assets [4]: Zenodo record `19498269`; archive 71,062 bytes.

C. Paper semantics and pinned implementation

Appendix C.2 defines reported index t and change time t^*: the first observation prefix with zero probability under the original environment, requiring the guard to coincide. D.2 and Listing D.3 describe selecting a historical observation and confirming it [1, pp. 15, 17, 19].

Boundary	Published task / interface [1]	Pinned public implementation [3]
Reported index	Score the historical index selected by the agent; show the chosen frame before confirmation.	Every `step()` sets `curr_frame = time-1`. Submission overwrites the prior selection before grading.
Early penalty	Score 0 when $t < t^* - 1$; full credit at $t^* - 1$ or t^*.	The corresponding raw too-early branch returns -1. No undocumented normalization is assumed.
Change target	First observation-prefix support violation; the guard must coincide with it.	`trigger_start_time` is assigned from a triggering condition. Equivalence to the paper's t^* is unverified.

Static mechanism. In `CDSliderEnvironment.step()`, time advances and `curr_frame` is assigned before dispatch. `choose_frame_5` then assigns 5, but the next `Submit choice` replaces it before `change_reward()` reads it [3, step()] [3, reward].

Source-derived reconstruction, not a new runtime observation. From saved time 7, declaration, selection and submission imply action-counter values 8, 9 and 10. Source flow therefore grades index **9 rather than the selected 5**. This does not recover the missing numerical return or certify the intended paper target.

Protocol qualification. The reset wrapper obtains an initial observation omitted by the pilot. The supplied agent submits immediately without frame selection [3]. All seven saved CD grids were identical; this alone does not determine original-environment support or validate guard time 6 as the paper's change time.

D. Minimal interpretation

The evidence establishes a discrepancy among the inspected public implementation, its selection protocol and the published definition. It does *not* establish a bug in the code used for the published experiments, invalid reported results, faithful official reproduction, a validated scoring repair, better world-model performance or a successful controlled-mutation study. The historical experiment remains `PHASE2_STOP`; an unqualified reproduction requires author clarification.

E. Focused research direction

How can environment-level evaluation preserve the semantics of a query across simulator dynamics, interaction protocols and reward implementations? An intended protocol or author-confirmed revision would provide a basis for later controlled evaluation. Any score-changing adapter would be separately labelled as an independent variant, with raw and adapted scores kept distinct. No scorer was patched, no adapter was implemented, and no further experiment is promised or authorized by this note.

References and exact code locations

[1] A. Warrier et al. *Benchmarking World-Model Learning with Environment-Level Queries.* ICML 2026, PMLR 306:133215–133246. Proceedings; C.2 (PDF p. 15), D.2 (p. 17), Listing D.3 (p. 19).

[2] BasisResearch/Autumn.cpp, exact revision on p. 1.

[3] BasisResearch/MARAProtocol, exact revision on p. 1. `concrete_envs.py`: step L362–422; reward L439–449; prompt/frames L451–489. `environment_interfaces.py`: Reset L56–67; Step L82–103; composite CD L273–295. `simple_wm_agent.py`: L259–271. Paths are under `python_examples/autumnbench/`.

[4] Warrier et al. *AutumnBench Public Benchmark*, Zenodo 19498269, CC BY 4.0. Source hashes and claim-level evidence are recorded in the accompanying claim check.